

7.1.1

Prove that for two arbitrary intervals X and Y at least one of the following cases holds:

- X and Y are disjoint and $X \rightarrow Y$; or
- X and Y are disjoint and $Y \rightarrow X$; or
- X and Y are overlapping intervals



Let $X = (x1, x2)$, $Y = (y1, y2)$, $x1 \rightarrow x2$, $y1 \rightarrow y2$. Then we have 6 possible cases. Lets brute-force:

- $x1 \rightarrow x2 \rightarrow y1 \rightarrow y2$: disjoint, $X \rightarrow Y$
- $x1 \rightarrow y1 \rightarrow x2 \rightarrow y2$: overlap
- $x1 \rightarrow y1 \rightarrow y2 \rightarrow x2$: overlap
- $y1 \rightarrow x1 \rightarrow x2 \rightarrow y2$: overlap
- $y1 \rightarrow x1 \rightarrow y2 \rightarrow x2$: overlap
- $y1 \rightarrow y2 \rightarrow x1 \rightarrow x2$: disjoint, $Y \rightarrow X$



7.1.2

Prove that $\rightarrow$ (precedence) binary relation on intervals is a partial order that:

- Irreflexive. Never true that $X \rightarrow X$.
- Antisymmetric. If $X \rightarrow Y$ then not true $Y \rightarrow X$.
- Transitive. If $X \rightarrow Y$ and $Y \rightarrow Z$ then $X \rightarrow Z$.

Please also prove that precedence is indeed a partial order: provide an example with $X \neq Y$ where $X \rightarrow Y$ false and $Y \rightarrow X$ is also false.



Let $X = (x1, x2)$, $Y = (y1, y2)$, $x1 \rightarrow x2$, $y1 \rightarrow y2$.

- Irreflexive because any interval overlaps itself.
- Antisymmetry follows from the asymmetry of the primordial order.
- Transitivity follows from transitivity of the primordial order.
- Partial because never true for overlapping intervals.

